

✓ SOP #8 – ATI (Analysis to Improve) for AHUs & RTUs

Goal: Enhance ventilation performance, reduce energy waste, and improve air quality through smart upgrades

1. Key Optimization Metrics

Metric	Ideal Range or Target	Optimization Action
ΔT (Supply – Return Air)	17–23°F cooling / 25–35°F heating	Clean coils, verify damper operation
Filter Pressure Drop	< 0.5" w.c. new, 1.0" max	Add filter differential gauges and change SOP
Economizer Functionality	Operates < 70°F OAT	Add CO ₂ sensors to trigger demand ventilation
Unit Runtime Efficiency	> 85% of set demand	Install VFD on supply fan or optimize BAS logic
Air Quality Index (Indoor)	0–50 AQI	Adjust outside air intake and monitor VOCs

3. Legal, Safety & Compliance Considerations

ASHRAE 62.1 & 90.1: Ventilation and energy standards
 NFPA 90A: Fire protection for AHUs and ductwork
 OSHA 1910.147 (LOTO): Required during servicing of fan/motor units
 EPA Indoor AirPLUS: For schools, hospitals, or LEED facilities
 NJ State Clean Air Regulations: Mandated for facilities over 10,000 sq ft

2. Strategic Improvements

Install CO₂ Demand-Controlled Ventilation (DCV): Reduces over-ventilation
 Use VFDs on Supply/Return Fans: Match load and reduce energy use
 UV Lights or Bipolar Ionization: Improves air cleanliness and coil hygiene
 Economizer Optimization: Set up mixed air logic with dewpoint control
 Enhanced Scheduling via BMS: Avoids unnecessary runtime